RAJALAKSHMI
INSTITUTE OF
TECHNOLOGY

BELIEVE IN THE POSSIBILITIES

AN AUTONOMOUS INSTITUTION

B.E. COMPUTER SCIENCE AND ENGINEERING

CURRICULUM FOR SEMESTER I TO VIII

REGULATIONS 2023

RAJALAKSHMI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Kuthambakkam, Chennai 600124

RAJALAKSHMI INSTITUTE OF TECHNOLOGY, CHENNAI
An Autonomous Institution, Affiliated to Anna University, Chennai

REGULATIONS 2023
CHOICE BASED CREDIT SYSTEM

B.E. COMPUTER SCIENCE AND ENGINEERING

I VISION OF THE DEPARTMENT

To establish a pioneering presence in the domain of Computer Science and Engineering by delivering excellence in technical learning that nurtures a culture of innovation, research and competent professionalism.

II MISSION OF THE DEPARTMENT

- To empower the next generation of Computer Science and Engineering professionals attuned to industry evolution and national progress.
- To drive a continuous advancement in teaching and learning standards in the field of Computer Science and Engineering by fostering an enriching research milieu.
- To forge a strong and symbiotic relationship with the industry, facilitating dynamic interactions that bridge academia and real-world applications, nurturing innovationand productive collaboration.

III PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates can

- Pursue higher education and research or have a successful career in industriesassociated with Computer Science and Engineering.
- Adapt the emerging technological changes for the global social benefits.
- Become entrepreneurs in the field of Computer Science and Engineering, inculcating Research and Innovation.

IV PROGRAM OUTCOMES (POs)

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needswith appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge andresearch methods, including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledgeto assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and Sustainability:** Understand the impact of the professional engineering solutions to societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a memberor leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities withthe engineering community and with society at large, such as, being able tocomprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project Management and Finance:** Demonstrate knowledge and understanding ofthe engineering and management principles and apply these to one's own work, asa member and leader in a team, to manage projects and in multidisciplinaryenvironments.
12. **Lifelong Learning:** Recognize the need for, and have the preparation and ability toengage in independent and lifelong learning in the broadest context of technologicalchange.

V PROGRAM SPECIFIC OUTCOMES (PSOs)

The students will be able to

- Design and programming skills to build and automate business solutions usingcutting edge technologies.
- Strengthen the theoretical foundation leading to excellence and excitement towardsresearch, to provide elegant solutions to complex problems.
- Work effectively with various engineering fields as a team to design, build anddevelop system applications.

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REGULATIONS 2023
CHOICE BASED CREDIT SYSTEM

B.E. COMPUTER SCIENCE AND ENGINEERING

CURRICULUM FOR SEMESTER I TO VIII

SEMESTER I

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
	IP23111	Induction Programme	-	-	-	-	-	0
THEORY COURSES								
1	HS23111	Communicative English	HSMC	3	0	0	3	3
2	CY23111	Engineering Chemistry	BSC	3	0	0	3	3
3	MA23111	Matrices and Calculus	BSC	3	1	0	4	4
4	GE23111	Problem Solving and C Programming	ESC	3	0	0	3	3
5	GE23112	தமிழர் மரபு /Heritage of Tamils	HSMC	1	0	0	1	0
LABORATORARY ORIENTED THEORY COURSE								
6	GE23131	Engineering Graphics	ESC	2	0	4	6	4
LABORATORARY COURSES								
7	CY23121	Chemistry Laboratory	BSC	0	0	2	2	1
8	GE23121	Problem Solving and C Programming Laboratory	ESC	0	0	2	2	1
TOTAL				**15**	**1**	**10**	**26**	**19**

SEMESTER II

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSES								
1	HS23211	Professional English	HSMC	2	0	0	2	2
2	MA23211	Statistics and Numerical Methods	BSC	3	1	0	4	4
3	PH23211	Physics for Information Science	BSC	3	0	0	3	3
4	AD23211	Python for Data science	ESC	3	0	0	3	3
5	GE23211	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	3
6	GE23213	தமிழரும் தொழில் நுட்பமும் /Tamils and Technology	HSMC	1	0	0	1	0
LABORATORARY COURSES								
7	PH23221	Physics Laboratory	BSC	0	0	2	2	1
8	AD23221	Python for Data Science Laboratory	ESC	0	0	2	2	1
9	GE23222	Engineering Practices Laboratory	ESC	0	0	2	2	1
10	GE23221	Communication Laboratory / Foreign Language	EEC	0	0	2	2	1
		NCC/Service Club Credit Course Level 1#		2	0	0	2	2#
		TOTAL						**19**

NCC Credit Course Level 1 is offered for NCC students only. The grades earned by the students will be recorded in the Mark Sheet, however the same shall not be considered for the computation of CGPA.

SEMESTER III

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSES								
1	GE23311	Environmental Science and Sustainability	BSC	2	0	0	2	2
2	MA23311	Discrete Mathematics	BSC	3	1	0	4	4
3	AL23311	Artificial Intelligence	PCC	3	0	0	3	3
4	CS23311	Data Structures	PCC	3	0	0	3	3
5	CS23312	Object Oriented Programming	PCC	3	0	0	3	3
LABORATORARY ORIENTED THEORY COURSE								
6	EC23331	Digital Principles and Computer Organization	ESC	3	0	2	5	4
LABORATORARY COURSES								
7	CS23321	Data Structures Laboratory	PCC	0	0	2	2	1
8	CS23322	Object Oriented Programming Laboratory	PCC	0	0	2	2	1
INDUSTRY ORIENTED COURSE								
9	CS23IC1	Design Thinking for Software Engineers	EEC	1	-	-	1	1
		TOTAL						**22**

SEMESTER IV

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSES								
1	CS23411	Database Management Systems	PCC	3	0	0	3	3
2	CS23412	Operating Systems	PCC	3	0	0	3	3
3	CS23413	Theory of Computation	PCC	3	0	0	3	3
4	CS23414	Software Development Practices	PCC	3	0	0	3	3
LABORATORARY ORIENTED THEORY COURSE								
5	AL23432	Machine Learning Techniques	PCC	3	0	2	5	4
6	CS23431	Design and Analysis of Algorithms	PCC	3	0	2	5	4
LABORATORARY COURSES								
7	CS23421	Database Management Systems Laboratory	PCC	0	0	2	2	1
8	CS23422	Operating Systems Laboratory	PCC	0	0	2	2	1
INDUSTRY ORIENTED COURSE								
9	CS23IC2	Visualization Tools	EEC	1	-	-	1	1
		NCC /Service Club Credit Course Level 2#		3	0	0	3	3#
		TOTAL						**23**

NCC Credit Course Level 2 is offered for NCC students and Service Club students only.

Thegrades earned by the students will be recorded in the Mark Sheet, however the same shall not be considered for the computation of CGPA.

SEMESTER V

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSE								
1	CS23511	Computer Networks	PCC	3	0	0	3	3
2	CS23512	Compiler Design	PCC	3	0	0	3	3
3	CS23513	Cryptography and Cyber Security	PCC	3	0	0	3	3
4		Professional Elective I	PEC	-	-	-	-	3
5		Professional Elective II	PEC	-	-	-	-	3
6		Mandatory Course-I&	MC	3	0	0	3	0
LABORATORARY ORIENTED THEORY COURSE								
7	AD23532	Data Exploration and Visualization	PCC	3	0	2	5	4
LABORATORARY COURSES								
8	CS23521	Computer Networks Laboratory	PCC	0	0	2	2	1
INDUSTRY ORIENTED COURSE								
9	CS23IC3	Industry Oriented Course –III	EEC	1	-	-	-	1
		TOTAL						**21**

&Mandatory Course-I is a Non-credit Course (Student shall select one course from the list given under Mandatory Course-I)

SEMESTER VI

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSE								
1		Professional Elective III	PEC	-	-	-	-	3
2		Professional Elective IV	PEC	-	-	-	-	3
3		Open Elective – I*	OEC	3	0	0	3	3
4		Open Elective – II*	OEC	3	0	0	3	3
5		Mandatory Course-II&	MC	3	0	0	3	0
LABORATORARY ORIENTED THEORY COURSE								
6	CS23631	Object Oriented Software Engineering	PCC	3	0	2	5	4
7	EC23631	Embedded Systems and IOT	ESC	3	0	2	5	4
LABORATORARY COURSE								
8	CS23621	Mini Project	EEC	0	0	4	4	2
		NCC /Service Club Credit Course Level 3#		3	0	0	3	3#
		TOTAL						22

*Open Elective – I and II Shall be chosen from the list of open electives offered by other Programmes.

&Mandatory Course-II is a Non-credit Course (Student shall select one course from the list given under Mandatory Course-II)

#NCC Credit Course level 3 is offered for NCC students only. The grades earned by the students will be recorded in the Mark Sheet, however the same shall not be considered for the computation of CGPA.

SEMESTER VII

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSE								
1	GE23711	Human Values and Ethics	HSMC	2	0	0	2	2
2		Professional Elective V	HSMC	3	0	0	3	3
3		Professional Elective VI	PEC	-	-	-	-	3
4		Elective – Management $	PEC	-	-	-	-	3
LAB ORIENTED THEORY COURSE								
5	CS23731	Cloud Computing	PCC	3	0	2	5	4
6	AD23731	Deep Learning Techniques	PCC	3	0	2	5	4
LABORATORARY COURSE								
7	CS23721	Internship/Certification Course	EEC	-	-	-	-	2
		TOTAL						21

$ Elective –Management shall be chosen from the list of elective management courses.

SEMESTER VIII

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY COURSE								
1		Open Elective – III **	OEC	3	0	0	3	3
LABORATORARY COURSE								
2	CS23821	Project Work	EEC	0	0	20	20	10
		TOTAL		3	0	20	23	**13**

**Open Elective III- Shall be chosen from the list of open electives offered by other Programmes.

TOTAL CREDITS: 160

ELECTIVE – MANAGEMENT COURSES

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	GE23711	Engineering Economics and Financial Accounting	HSMC	3	0	0	3	3
2	GE23712	Human Resource Management	HSMC	3	0	0	3	3
3	GE23713	Knowledge Management	HSMC	3	0	0	3	3
4	GE23714	Principles of Management	HSMC	3	0	0	3	3
5	GE23715	Software Project Management	HSMC	3	0	0	3	3
6	GE23716	Total Quality Management	HSMC	3	0	0	3	3
7	GE23717	Management Information System	HSMC	3	0	0	3	3

MANDATORY COURSES I

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	MX23511	Disaster Risk Reduction and Management	MC	3	0	0	3	0
2	MX23512	Elements of Literature	MC	3	0	0	3	0
3	MX23513	Film Appreciation	MC	3	0	0	3	0
4	MX23514	Introduction to Women and GenderStudies	MC	3	0	0	3	0

MANDATORY COURSES II

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	MX23611	History of Science and Technologyin India	MC	3	0	0	3	0
2	MX23612	Industrial Safety	MC	3	0	0	3	0
3	MX23613	State, Nation Building and Politics inIndia	MC	3	0	0	3	0
4	MX23614	Well Being with Traditional Practices - Yoga, Ayurveda and Siddha	MC	3	0	0	3	0

PROFESSIONAL ELECTIVE COURSES: VERTICALS

Sl. No.	Vertical 1	Vertical 2	Vertical 3	Vertical 4	Vertical 5	Vertical 6	Vertical 7
	Data Science	**Cloud Computing**	**Emerging Technologies**	**Artificial Intelligence**	**Full Stack Development**	**Supply Chain Management**	**Cyber Security and Data Privacy**
1	Introduction to Big Data Analytics	Cloud Solution Architecture	Cloud Solution Architecture	Generative AI	Cloud Services Management	Industry 5.0	Security and Privacy in Cloud
2	Computer Vision	Configuration Management	Game Development	Cognitive Science	Mobile App Development	Supply Chain Management	Cryptocurrency and Block chain Technologies
3	Healthcare Analytics	Cloud Virtualization	Drone Technology	Responsive AI	Agile and DevOps	Planning in Logistics	Digital and Mobile Forensics
4	Exploratory Data Analysis	Container Orchestration	Robotic Process Automation	Game Theory	MLOps	Warehouse Automation	Engineering Secure Software Systems
5	Image and Video Analytics	Cloud Services Management	Autonomous Vehicle	Knowledge Engineering	Software Testing and Automation	Supply Chain for Manufacturing	Ethical Hacking
6	Recommender Systems	Security and Privacy in Cloud	Cryptocurrency and Blockchain Technologies	Reinforcement Learning	UI and UX Design	Supply Chain Information System	Modern Cryptography
7	Text and Speech Analysis	Storage Technologies	Augmented Reality/Virtual Reality	Computer Vision	Web Application Security	Sustainable Inventory Management	Network Security
8	Data Mining and Warehousing	Software Defined Networks	Connected Vehicles	Text and Speech Analysis	Web Technologies	Supply Chain Analytics	Social Network Security

Registration of Professional Elective Courses from Verticals:

Professional Elective Courses will be registered in Semesters V, VI and VII. These courses are listed in groups called verticals that represent a particular area of specialisation / diversified group. Students are permitted to choose all the Professional Electives from a particular vertical or from different verticals. Further, only one Professional Elective course shall be chosen in a semester horizontally (row-wise). However, two courses are permitted from the same row, provided one course is enrolled in Semester V and another in semester VI. The registration of courses for B.E./B.Tech (Honours) or Minor degree shall be done from Semester V to VIII. The procedure for registration of courses explained above shall be followed for the courses of B.E/B.Tech (Honours) or Minor degree also.

PROFESSIONAL ELECTIVE COURSES: VERTICALS

VERTICAL 1: DATA SCIENCE

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V11	Introduction to Big Data Analytics	PEC	2	0	2	4	3
2	CS23V12	Computer Vision	PEC	2	0	2	4	3
3	CS23V13	Healthcare Analytics	PEC	2	0	2	4	3
4	CS23V14	Exploratory Data Analysis	PEC	2	0	2	4	3
5	CS23V15	Image and Video Analytics	PEC	2	0	2	4	3
6	CS23V16	Recommender Systems	PEC	2	0	2	4	3
7	CS23V17	Text and Speech Analysis	PEC	2	0	2	4	3
8	CS23V18	Data Mining and Warehousing	PEC	2	0	2	4	3

VERTICAL 2: CLOUD COMPUTING

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V21	Cloud Solution Architecture	PEC	2	0	2	4	3
2	CS23V22	Configuration Management	PEC	2	0	2	4	3
3	CS23V23	Cloud Virtualization	PEC	2	0	2	4	3
4	CS23V24	Container Orchestration	PEC	2	0	2	4	3
5	CS23V25	Cloud Services Management	PEC	2	0	2	4	3
6	CS23V26	Security and Privacy in Cloud	PEC	2	0	2	4	3
7	CS23V27	Storage Technologies	PEC	2	0	2	4	3
8	CS23V28	Software Defined Networks	PEC	2	0	2	4	3

VERTICAL 3: EMERGING TECHNOLOGIES

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V21	Cloud Solution Architecture	PEC	2	0	2	4	3
2	CS23V31	Game Development	PEC	2	0	2	4	3
3	CS23V32	Drone Technology	PEC	2	0	2	4	3
4	CS23V33	Robotic Process Automation	PEC	2	0	2	4	3
5	CS23V34	Autonomous Vehicle	PEC	2	0	2	4	3
6	CS23V35	Cryptocurrency and Blockchain Technologies	PEC	2	0	2	4	3
7	CS23V36	Augmented Reality/Virtual Reality	PEC	2	0	2	4	3
8	CS23V37	Connected Vehicles	PEC	2	0	2	4	3

VERTICAL 4: ARTIFICIAL INTELLIGENCE

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V41	Generative AI	PEC	2	0	2	4	3
2	CS23V42	Cognitive Science	PEC	2	0	2	4	3
3	CS23V43	Responsive AI	PEC	2	0	2	4	3
4	CS23V44	Game Theory	PEC	2	0	2	4	3
5	CS23V45	Knowledge Engineering	PEC	2	0	2	4	3
6	CS23V46	Reinforcement Learning	PEC	2	0	2	4	3
7	CS23V12	Computer Vision	PEC	2	0	2	4	3
8	CS23V17	Text and Speech Analysis	PEC	2	0	2	4	3

VERTICAL 5: FULL STACK DEVELOPMENT

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V25	Cloud Services Management	PEC	2	0	2	4	3
2	CS23V51	Mobile App Development	PEC	2	0	2	4	3
3	CS23V52	Agile and DevOps	PEC	2	0	2	4	3
4	CS23V53	MLOps	PEC	2	0	2	4	3
5	CS23V54	Software Testing and Automation	PEC	2	0	2	4	3
6	CS23V55	UI and UX Design	PEC	2	0	2	4	3
7	CS23V56	Web Application Security	PEC	2	0	2	4	3
8	CS23V57	Web Technologies	PEC	2	0	2	4	3

VERTICAL 6: SUPPLY CHAIN MANAGEMENT

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V61	Industry 5.0	PEC	2	0	2	4	3
2	CS23V62	Supply Chain Management	PEC	2	0	2	4	3
3	CS23V63	Planning in Logistics	PEC	2	0	2	4	3
4	CS23V64	Warehouse Automation	PEC	2	0	2	4	3
5	CS23V65	Supply Chain for Manufacturing	PEC	2	0	2	4	3
6	CS23V66	Supply Chain Information System	PEC	2	0	2	4	3
7	CS23V67	Sustainable Inventory Management	PEC	2	0	2	4	3
8	CS23V68	Supply Chain Analytics	PEC	2	0	2	4	3

VERTICAL 7: CYBER SECURITY AND DATA PRIVACY

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	CS23V26	Security and Privacy in Cloud	PEC	2	0	2	4	3
2	CS23V35	Cryptocurrency and Blockchain Technologies	PEC	2	0	2	4	3
3	CS23V71	Digital and Mobile Forensics	PEC	2	0	2	4	3
4	CS23V72	Engineering Secure Software Systems	PEC	2	0	2	4	3
5	CS23V73	Ethical Hacking	PEC	2	0	2	4	3
6	CS23V74	Modern Cryptography	PEC	2	0	2	4	3
7	CS23V75	Network Security	PEC	2	0	2	4	3
8	CS23V76	Social Network Security	PEC	2	0	2	4	3

OPEN ELECTIVES

(Students shall choose the open elective courses, such that the course contents are not similar to any other course contents/title under other course categories).

OPEN ELECTIVES – I

Sl. No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
Course Offered By Other Departments								
1	023EE11	Space Engineering	OEC	3	0	0	3	3
2	023EE12	IT in Agricultural and Fisheries System	OEC	3	0	0	3	3
3	023EC11	Fundamentals of VLSI	OEC	3	0	0	3	3
4	023MA11	Probability and Statistics for Data Analytics	OEC	3	0	0	3	3
5	023ME11	Foundation of Robotics	OEC	3	0	0	3	3
Course Offered To Other Departments								
1	023CS11	Introduction to Cloud Computing	OEC	3	0	0	3	3

OPEN ELECTIVES – II

Sl. No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
Course Offered By Other Departments								
1	023EE21	Space Science	OEC	3	0	0	3	3
2	023EC21	Wearable Devices and Applications	OEC	3	0	0	3	3
3	023EC22	Introduction to IOT	OEC	3	0	0	3	3
4	023MA21	Optimization Techniques	OEC	3	0	0	3	3
5	023ME21	Foundation of Mechatronics	OEC	3	0	0	3	3
Course Offered To Other Departments								
1	023CS21	Introduction to Cyber Security	OEC	3	0	0	3	3

OPEN ELECTIVES – III

Sl. No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
Course Offered By Other Departments								
1	023AL31	Introduction to Block Chain and Applications	OEC	3	0	0	3	3
2	023EE31	Fundamentals of Drone Technologies	OEC	3	0	0	3	3
3	023EE32	Batteries and Management System	OEC	3	0	0	3	3
4	023EC31	Medical Electronics	OEC	3	0	0	3	3
5	023MA31	Multivariate Data Analysis	OEC	3	0	0	3	3
6	023ME31	Introduction to PLC Programming	OEC	3	0	0	3	3
Course Offered To Other Departments								
1	023CS31	Augmented Reality/ Virtual Reality/ Extended Reality	OEC	3	0	0	3	3

SUMMARY

Name of the Programme: B.E. Computer Science and Engineering										
Sl.No.	Subject Area	Credits per Semester								Total Credits
		I	II	III	IV	V	VI	VII	VIII	
1	HSMC	3	2					5		10
2	BSC	8	8	6						22
3	ESC	8	8	4			3			23
4	PCC			11	22	14	5	8		60
5	PEC					6	6	6		18
6	OEC						6		3	9
7	EEC		1	1	1	1	2	2	10	18
8	Non-Credit /(Mandatory)					√	√			
	Total	19	19	22	23	21	22	21	13	160

ENROLLMENT FOR B.E. / B. TECH. (HONOURS) / MINOR DEGREE (OPTIONAL)

A student can also optionally register for additional courses (18 credits) and become eligible for the award of B.E. / B. Tech. (Honours) or Minor Degree.

For B.E. / B. Tech. (Honours), a student shall register for the additional courses (18 credits) from semester V onwards. These courses shall be from the same vertical or a combination of different verticals of the same programme of study only.

For minor degree, a student shall register for the additional courses (18 credits) from semester V onwards. All these courses have to be in a particular vertical from any one of the other programmes, Moreover, for minor degree the student can register for courses from any one of the following verticals also.

VERTICALS FOR MINOR DEGREE
(In addition to all the verticals of other programmes)

Sl. No.	Vertical 1	Vertical 2	Vertical 3	Vertical 4
	Fintech and Block Chain	**Entrepreneurship**	**Business Data Analytics**	**IOT**
1	Introduction to Fintech	Foundations of Entrepreneurship	Statistics for Management	Industrial IOT
2	Introduction to Block chain and its Applications	Team Building & Leadership Management for Business	Datamining for Business Intelligence	IOT Processor
3	Financial Management	Creativity & Innovation in Entrepreneurship	Human Resource Analytics	IOT based System Design
4	Banking, Financial Services and Insurance	Principles of Marketing Management for Business	Marketing and Social Media Web Analytics	Sensor Technologies and IOT
5	Fundamentals of Investment	Human Resource Management for Entrepreneurs	Operation and Supply Chain Analytics	IOT Protocols
6	Fintech Analytics	Financing New Business Ventures	Financial Analytics	IOT Applications

(choice of courses for Minor degree is to be made from any one vertical of other programmes or from anyone of the following verticals)

VERTICAL 1: FINTECH AND BLOCK CHAIN

Sl.No.	Course Code	Course Title	Category	Periods Per Week L	T	P	Total Contact Periods	Credits
1	CS23M01	Introduction to Fintech	PEC	3	0	0	3	3
2	CS23M02	Introduction to Blockchain and its Applications	PEC	3	0	0	3	3
3	CS23M03	Financial Management	PEC	3	0	0	3	3
4	CS23M04	Banking, Financial Services and Insurance	PEC	3	0	0	3	3
5	CS23M05	Fundamentals of Investment	PEC	3	0	0	3	3
6	CS23M06	Fintech Analytics	PEC	3	0	0	3	3

VERTICAL 2: ENTREPRENEURSHIP

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	ME23M01	Foundations of Entrepreneurship	PEC	3	0	0	3	3
2	ME23M02	Team Building & Leadership Management for Business	PEC	3	0	0	3	3
3	ME23M03	Creativity & Innovation in Entrepreneurship	PEC	3	0	0	3	3
4	ME23M04	Principles of Marketing Management for Business	PEC	3	0	0	3	3
5	ME23M05	Human Resource Management for Entrepreneurs	PEC	3	0	0	3	3
6	ME23M06	Financing New Business Ventures	PEC	3	0	0	3	3

VERTICAL 3: BUSINESS DATA ANALYTICS

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	AD23M01	Statistics for Management	PEC	3	0	0	3	3
2	AD23M02	Data Mining for Business Intelligence	PEC	3	0	0	3	3
3	AD23M03	Human Resource Analytics	PEC	3	0	0	3	3
4	AD23M04	Marketing and Social Media Web Analytics	PEC	3	0	0	3	3
5	AD23M05	Operation and Supply Chain Analytics	PEC	3	0	0	3	3
6	AD23M06	Financial Analytics	PEC	3	0	0	3	3

VERTICAL 4: IOT

Sl.No.	Course Code	Course Title	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	EC23M01	Industrial IOT	PEC	3	0	0	3	3
2	EC23M02	IOT Processor	PEC	3	0	0	3	3
3	EC23M03	IOT Based System Design	PEC	3	0	0	3	3
4	EC23M04	Sensor Technologies and IOT	PEC	3	0	0	3	3
5	EC23M05	IOT Protocols	PEC	3	0	0	3	3
6	EC23M06	IOT Applications	PEC	3	0	0	3	3